Anexo 5: Reporte de Procesamiento de Líneas Base GNSS – Vértice GPS-02

Presenta el informe técnico de procesamiento estático de las líneas base GNSS vinculadas al vértice GPS-02 a partir de las estaciones de operación continua BUEN y CALI, validando la fijación de ambigüedades de fase (Phase Fixed) y las precisiones vectoriales obtenidas con el software Leica Infinity.

Leica Geosystems AG
Heinrich Wild Strasse
CH-9435 Heerbrugg
St. Gallen, Switzerland

Phone: + 41 71 727 3131
Fax: + 41 71 727 4674

- when it has to be right

GNSS Processing Report - Summary

Report created: 06/10/2026 19:05:25

Project Details

General		Customer Details	
Project Name:	Daniel Cortes	Customer Name:	-
Owner:	-	Contact Person:	-
Lead Surveyor:	-	Number:	-
Date Created:	06/10/2026 18:18:57	Email:	-
Last Accessed:	06/10/2026 18:18:57	Skype:	-
Application Software:	Infinity 3.0	Website:	-

Master Coordinate System	
Coordinate System Name:	MAGNA-SIRGAS_Origen-Nacional
Transformation Type:	Classical 3D
Residual Distribution:	None
Ellipsoid:	GRS 1980
Projection Type:	Transverse Mercator
Geoid Model:	Geocol2004
CSCS Model:	-

Path:	C:\Users\usuario\Documents\Leica Geosystems\Infinity\Projects\Daniel Cortes\Daniel Cortes.iprj
Size:	57,2 MB
Comments:	-

Baseline BUEN - GPS-02

Processing Parameters (13/09/2026 09:45:00 - 13/09/2026 13:59:24)

Data	Selected	Used	Comments
Cut-Off Angle:	10°	10°	
Frequency:	Automatic	L1/L2	
Sampling Rate:	Use All	15,00 sec	
Satellite System:	GPS/GLONASS/Galileo/Beidou	GPS/GLONASS	
Ephemeris Type:	Precise	Precise	
Antenna Calibration Set:	NGS Absolute	NGS Absolute	

Processing Strategy

Solution Type:	Phase Fixed	Phase Fixed
Solution Optimisation:	Automatic	Iono Minimised
Frequency to use in Iono Minimised:	Automatic	L1/L2
Tropospheric Model:	VMF with GPT2 model	VMF with GPT2 model
Ionospheric Model:	Automatic	Computed
Allow Widelane Fix:	Automatic	Automatic

General Settings

Min. Distance for Iono Minimised:	15 km
Possible Ambiguities Fix up to:	300 km
Min. Duration for Float Solution (static):	00:05:00

Results Baseline: BUEN - GPS-02

Acquisition

Start Time - End Time:	13/09/2026 09:45:12 - 13/09/2026 13:59:12
Duration:	04:14:00

Antennas

	Reference - BUEN	Rover - GPS-02

Receiver Name / SN:	TRIMBLE NETR9 / 5134K78077	V30 Plus / 11662288
Antenna Name / SN:	TRM59800.00 SCIS / 5138354298	HITV30PLUS / -
Carrier Offset:	-	-
Height Reading:	0,1240 m	1,7251 m
Antenna Height:	0,1240 m	1,7251 m

Coordinates

	Reference - BUEN	Rover - GPS-02		Reference - BUEN	Rover - GPS-02
Point Role:	Control	Fixed PP			
WGS84 Latitude:	3° 52' 55,29063" N	3° 29' 03,94759" N	Easting:	4.554.573,0135 m	4.644.751,6891 m
WGS84 Longitude:	77° 00' 37,51018" W	76° 11' 52,87146" W	Northing:	1.988.021,5479 m	1.943.637,4128 m
WGS84 Ellip. Height:	57,7645 m	1.535,4770 m	Ortho. Height:	39,2398 m	1.503,2711 m
WGS84 Cartesian X:	1.430.383,9669 m	1.519.184,6250 m			
WGS84 Cartesian Y:	-6.200.818,1398 m	-6.184.089,2448 m			
WGS84 Cartesian Z:	428.934,2670 m	385.148,9442 m			

Baseline Vector and Quality - WGS84

ΔLatitude:	-0° 23' 51,34304"	SD ΔLatitude:	0,0004 m
ΔLongitude:	0° 48' 44,63872"	SD ΔLongitude:	0,0004 m
ΔHeight:	1.477,7126 m	SD ΔHeight:	0,0011 m
ΔX:	88.800,6581 m	SD ΔX:	0,0005 m
ΔY:	16.728,8950 m	SD ΔY:	0,0011 m
ΔZ:	-43.785,3228 m	SD ΔZ:	0,0004 m
Slope Dist.:	100.411,9879 m	SD Slope Dist.:	0,0004 m

M0:	0,4925 m	CQ 1D:	0,0011 m
Q11:	0,00000090	CQ 2D:	0,0005 m
Q12:	-0,00000100	CQ 3D:	0,0012 m
Q22:	0,00000456		
Q13:	0,00000004		
Q23:	-0,00000035		
Q33:	0,00000058		

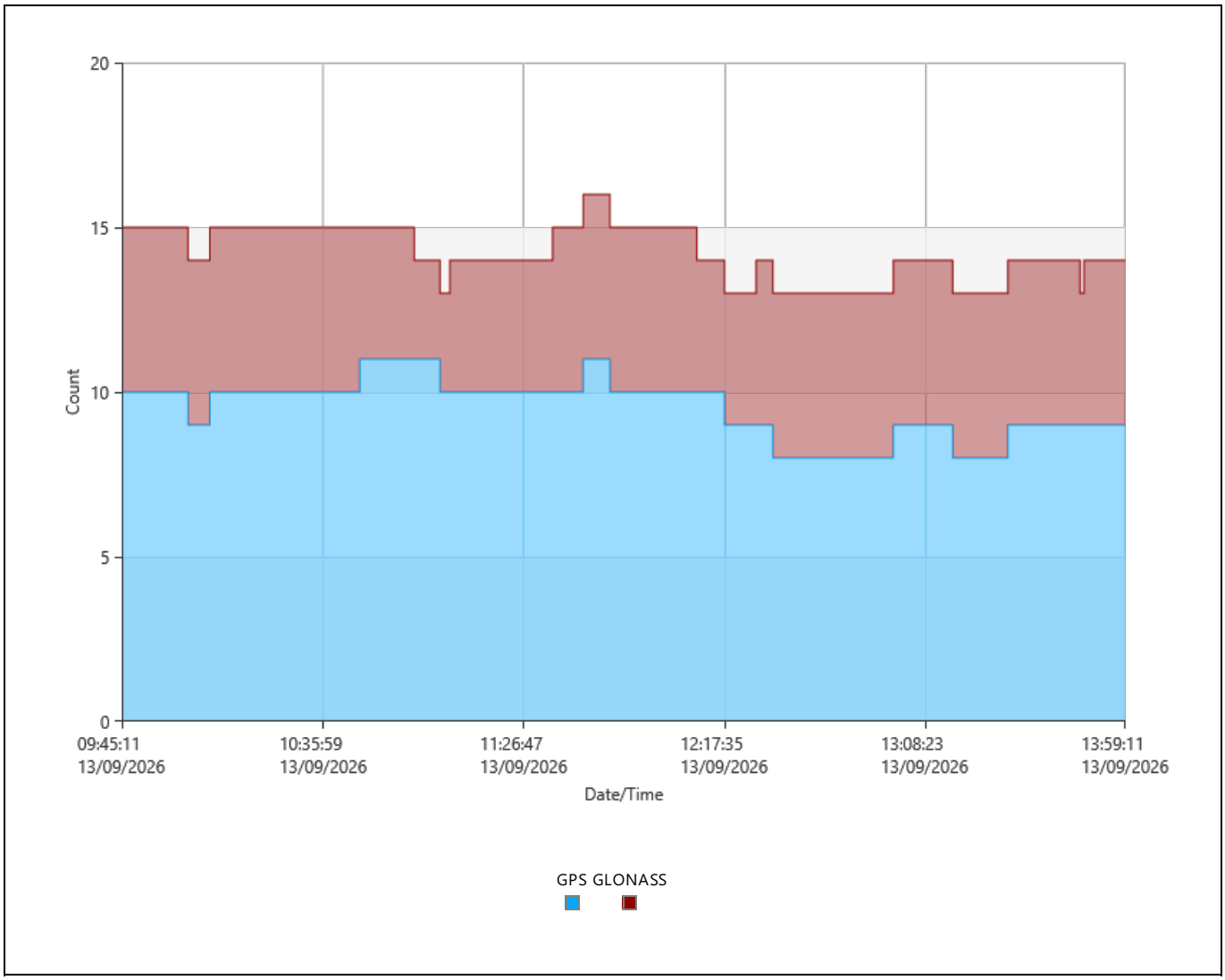
Frequency:	L1/L2	GDOP:	1,5 - 2,3	GPS SVs:	11/11
Solution Optimisation:	Iono Minimised	PDOP:	1,2 - 1,8	GLONASS SVs:	5/5
Solution Type:	Phase Fixed	HDOP:	0,6 - 0,9	Beidou SVs:	-
		VDOP:	1,0 - 1,7	Galileo SVs:	-
				QZSS SVs:	-

Ephemeris Type:	
GPS	Precise
GLONASS	Precise

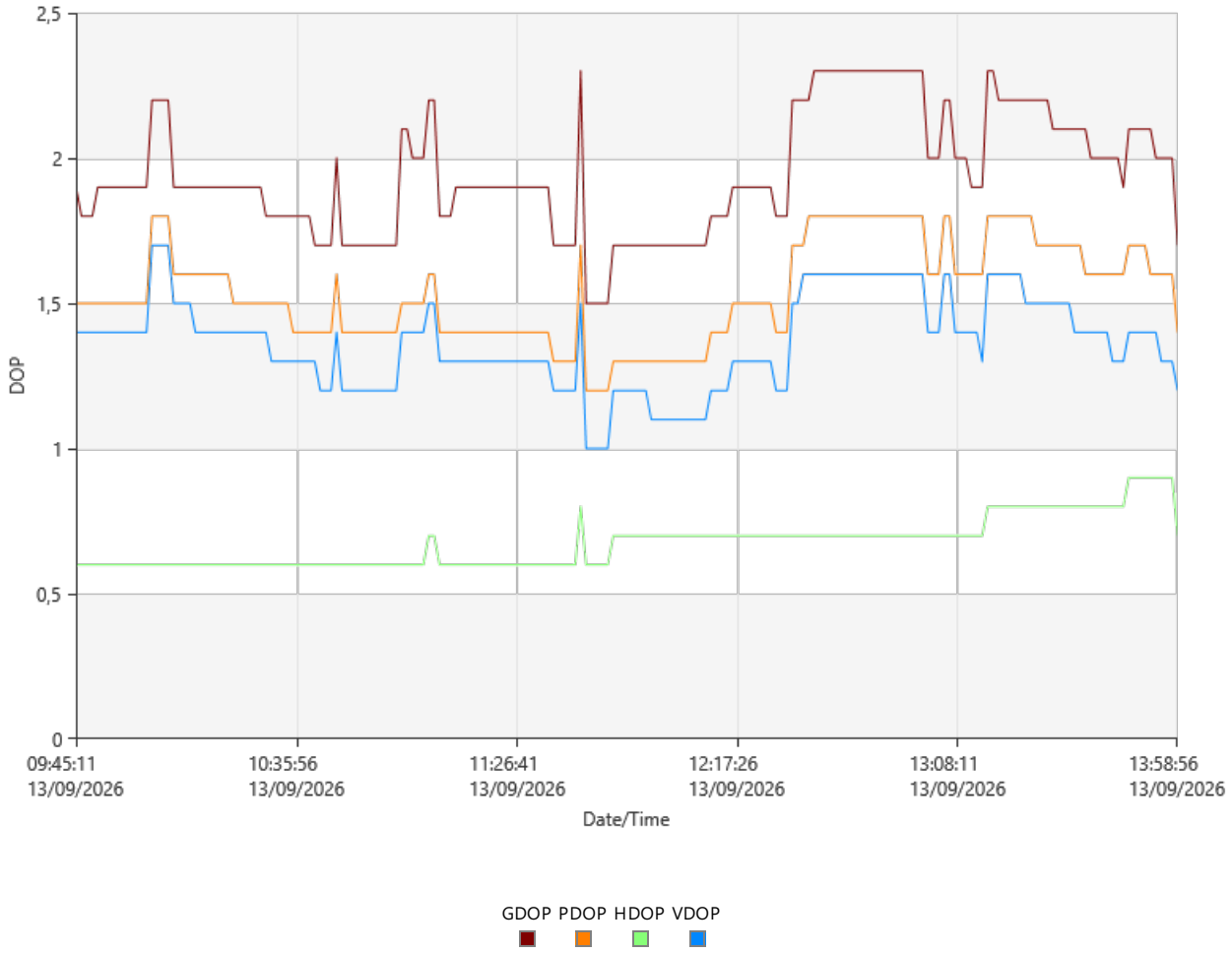
Processing Info (13/09/2026 09:45:00 - 13/09/2026 13:59:24)

Processed Date/Time:	06/10/2026 19:00:33

SVs Tracked


20
15
10
5
0
Count
09:45:11
13/09/2026
10:35:59
13/09/2026
11:26:47
13/09/2026
12:17:35
13/09/2026
13:08:23
13/09/2026
13:59:11
13/09/2026
Date/Time
GPS GLONASS

DOP



Ambiguity Statistics

Number of Ambiguities	GPS	GLONASS
Fixed	42	16
Total	60	26
Independently fixed	726	723
Possible independently fixed	901	901

Average time between independent fixes: 00:01:30

% of Epochs	GPS		GLONASS	
	L1 [%]	L2 [%]	L1 [%]	L2 [%]
Fixed	98,40	98,36	98,70	98,70
Not fixed	1,60	1,64	1,30	1,30
Not fixed - contradiction	0,00	0,00	0,00	0,00
Not fixed - missing phase	0,00	0,00	0,00	0,00

Status	From Epoch	To Epoch	Duration
Fixed	13/09/2026 09:45:12	13/09/2026 13:58:12	04:13:00
Not fixed	**13/09/2026 13:58:12**	**13/09/2026 13:59:12**	**00:01:00**

Baseline CALI - GPS-02

Processing Parameters (13/09/2026 09:45:00 - 13/09/2026 13:59:24)

Data	Selected	Used	Comments
Cut-Off Angle:	10°	10°	
Frequency:	Automatic	L1/L2/L5	
Sampling Rate:	Use All	15,00 sec	
Satellite System:	GPS/GLONASS/Galileo/Beidou	GPS/GLONASS	

Ephemeris Type:	Precise	Precise
Antenna Calibration Set:	NGS Absolute	NGS Absolute

Processing Strategy

Solution Type:	Phase Fixed	Phase Fixed
Solution Optimisation:	Automatic	Iono Minimised
Frequency to use in Iono Minimised:	Automatic	L1/L2/L5
Tropospheric Model:	VMF with GPT2 model	VMF with GPT2 model
Ionospheric Model:	Automatic	Computed
Allow Widelane Fix:	Automatic	Automatic

General Settings

Min. Distance for Iono Minimised:	15 km
Possible Ambiguities Fix up to:	300 km
Min. Duration for Float Solution (static):	00:05:00

Results Baseline: CALI - GPS-02

Acquisition

Start Time - End Time:	13/09/2026 09:45:12 - 13/09/2026 13:59:12
Duration:	04:14:00

Antennas

	Reference - CALI	Rover - GPS-02
Receiver Name / SN:	TRIMBLE NETR9 / 5145K79436	V30 Plus / 11662288
Antenna Name / SN:	TRM59800.00 SCIS / 5134354245	HITV30PLUS / -
Carrier Offset:	-	-
Height Reading:	0,1400 m	1,7251 m
Antenna Height:	0,1400 m	1,7251 m

Coordinates

	Reference - CALI	Rover - GPS-02		Reference - CALI	Rover - GPS-02
Point Role:	Control	Fixed PP			
WGS84 Latitude:	3° 22' 32,84048" N	3° 29' 03,95343" N	Easting:	4.607.499,9398 m	4.644.751,6056 m
WGS84 Longitude:	76° 31' 57,23035" W	76° 11' 52,87418" W	Northing:	1.931.743,8449 m	1.943.637,5927 m
WGS84 Ellip. Height:	1.027,4846 m	1.535,5400 m	Ortho. Height:	998,5097 m	1.503,3341 m
WGS84 Cartesian X:	1.483.100,0097 m	1.519.184,5559 m			
WGS84 Cartesian Y:	-6.193.060,1469 m	-6.184.089,3152 m			
WGS84 Cartesian Z:	373.124,3315 m	385.149,1272 m			

Baseline Vector and Quality - WGS84

ΔLatitude:	0° 06' 31,11296"	SD ΔLatitude:	0,0003 m
ΔLongitude:	0° 20' 04,35617"	SD ΔLongitude:	0,0003 m
ΔHeight:	508,0554 m	SD ΔHeight:	0,0008 m
ΔX:	36.084,5461 m	SD ΔX:	0,0003 m
ΔY:	8.970,8317 m	SD ΔY:	0,0008 m
ΔZ:	12.024,7957 m	SD ΔZ:	0,0003 m
Slope Dist.:	39.078,9714 m	SD Slope Dist.:	0,0003 m

M0:	0,4059 m	CQ 1D:	0,0008 m
Q11:	0,00000074	CQ 2D:	0,0004 m
Q12:	-0,00000078	CQ 3D:	0,0009 m
Q22:	0,00000356		
Q13:	0,00000005		
Q23:	-0,00000030		
Q33:	0,00000046		

Frequency:	L1/L2/L5	GDOP:	1,7 - 2,3	GPS SVs:	10/10
Solution Optimisation:	Iono Minimised	PDOP:	1,3 - 1,8	GLONASS SVs:	5/5

Solution Type:	Phase Fixed	HDOP:	0,6 - 0,9	Beidou SVs:	-
		VDOP:	1,1 - 1,7	Galileo SVs:	-
				QZSS SVs:	-

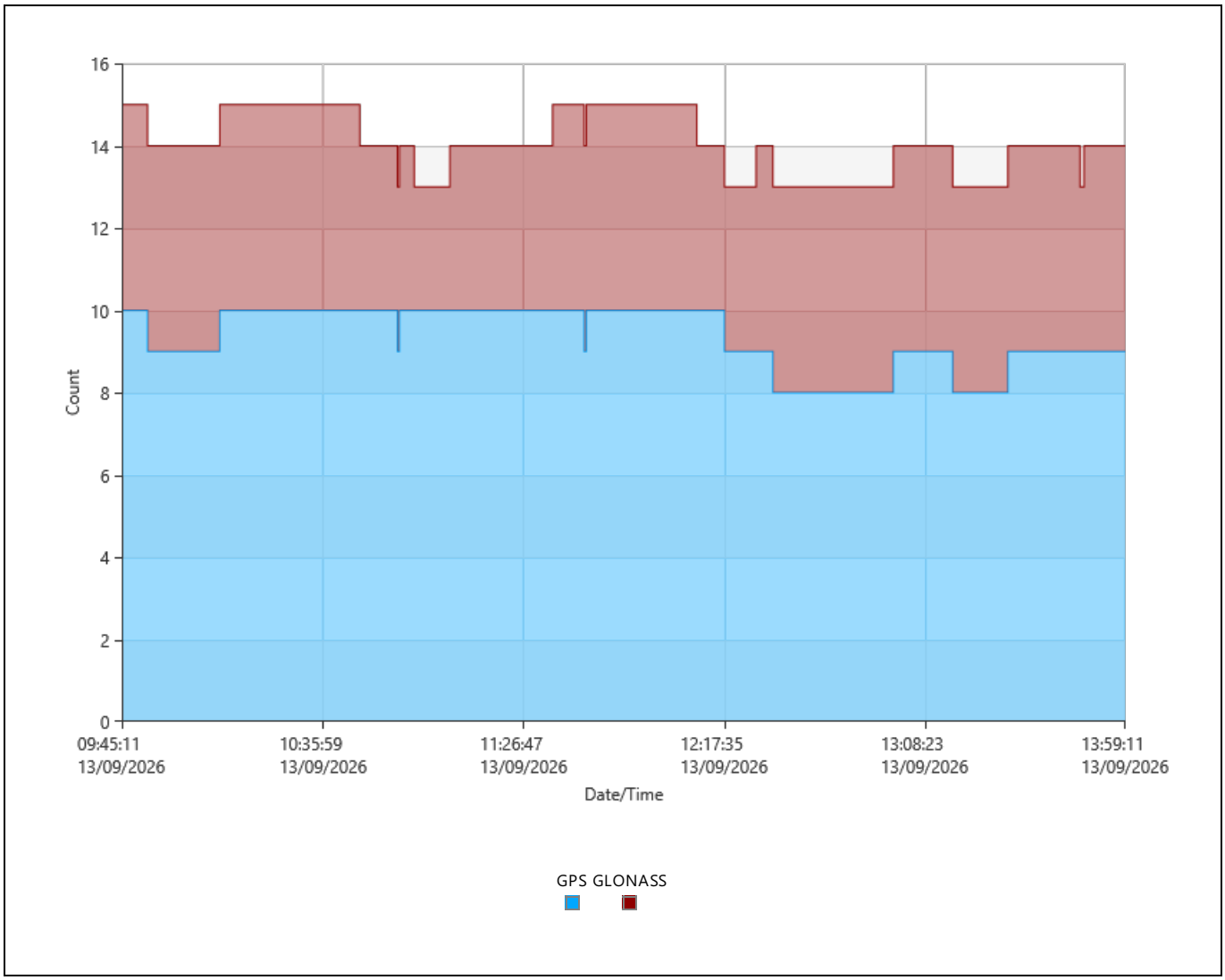
Ephemeris Type:

GPS	Precise
GLONASS	Precise

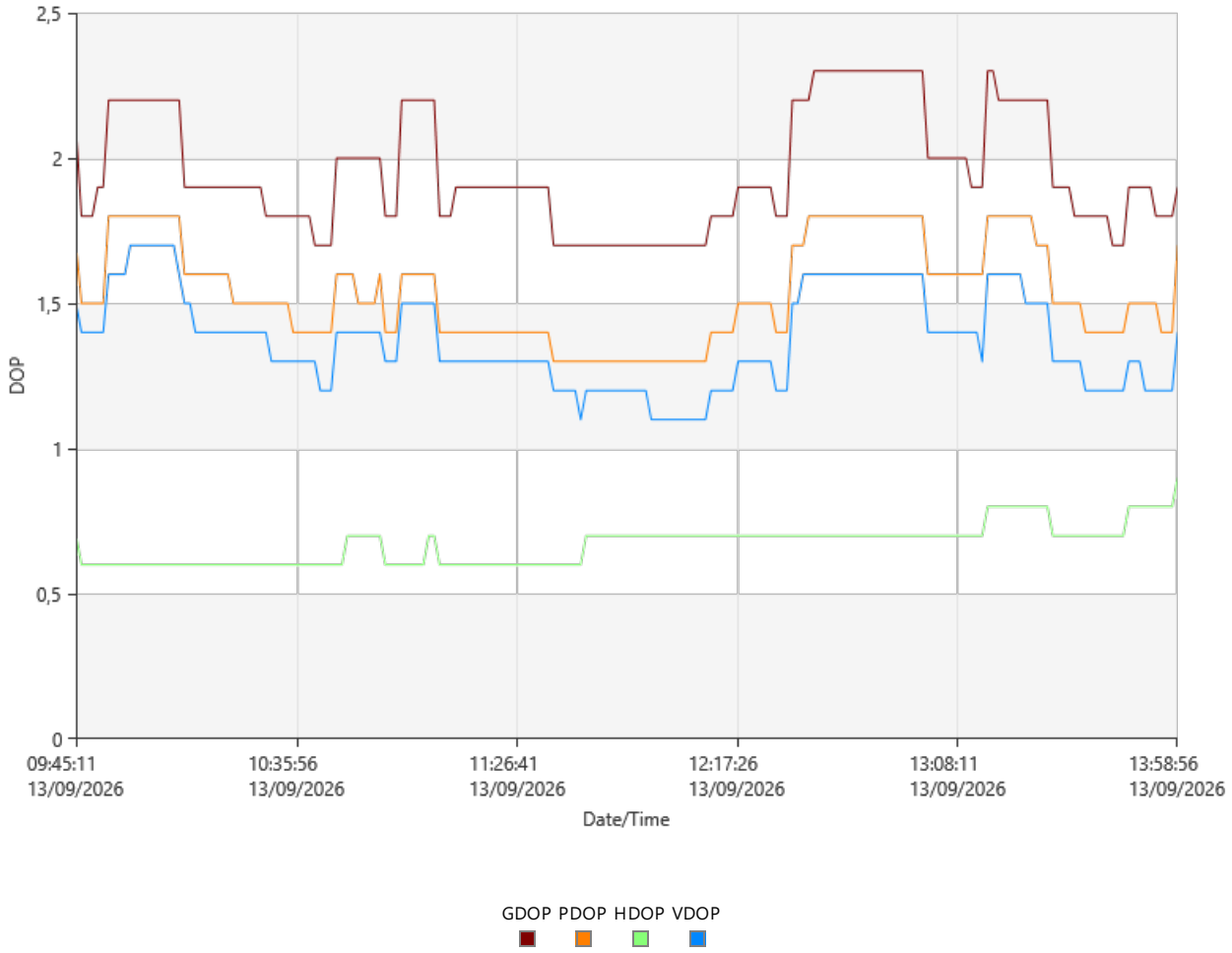
Processing Info (13/09/2026 09:45:00 - 13/09/2026 13:59:24)

Processed Date/Time: 06/10/2026 19:00:33

SVs Tracked



DOP



Ambiguity Statistics

Number of Ambiguities	GPS	GLONASS
Fixed	61	16
Total	64	26
Independently fixed	790	785
Possible independently fixed	818	818

Average time between independent fixes: 00:01:15

% of Epochs	GPS			GLONASS	
	L1 [%]	L2 [%]	L5 [%]	L1 [%]	L2 [%]
Fixed	99,99	99,93	100,00	98,70	98,70
Not fixed	0,01	0,07	0,00	1,30	1,30
Not fixed - contradiction	0,00	0,00	0,00	0,00	0,00
Not fixed - missing phase	0,00	0,00	0,00	0,00	0,00

Status	From Epoch	To Epoch	Duration
Fixed	13/09/2026 09:45:12	13/09/2026 13:59:12	04:14:00